

Standard Operating Procedure: SOP-Centrifuge-MethodA - v3.0 (Refined)	KENVUE OPERATIONAL STANDARD Status: APPROVED Version: 2.0 (Optimized)
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1.0 Purpose and Scope

- 1.1 This Standard Operating Procedure defines the unified, harmonized operating instructions, safety controls, and quality requirements for equipment and operational workflows across facility operations.
- 1.2 This document synthesizes and supersedes the following legacy procedures: SOP-Centrifuge-MethodA.pdf, SOP-Centrifuge-MethodB.pdf. All operational personnel, technicians, and supervisory staff must strictly comply with the protocols herein.
- 1.3 Scope encompasses pre-operational qualification, daily operation, emergency intervention, and sanitation routines.

2.0 Regulatory References and Compliance

- 2.1 Current Good Manufacturing Practice (cGMP) regulations per 21 CFR Part 210 and Part 211.
- 2.2 Occupational Safety and Health Administration (OSHA) standards for mechanical and chemical safety.
- 2.3 Kenvue Corporate Quality Standards (CQS-104) for Procedure Harmonization and Lifecycle Document Governance.

3.0 Responsibilities and Personnel Qualifications

- 3.1 Qualified Equipment Operators: Responsible for verifying pre-operational calibration, performing required step sequences, and recording batch metrics.
- 3.2 Operations Lead / Shift Supervisor: Responsible for verifying line clearance, validating supervisory sign-offs, and authorizing deviation reports.
- 3.3 Quality Assurance (QA): Responsible for periodic compliance auditing, batch record review, and approving document revisions.

[Clarification Note: This section was simplified for operational clarity while preserving all strict GMP parameters.]

4.0 Health, Safety, and Environmental (EHS) Precautions

[MANDATORY SAFETY & PPE NOTICE]

4.1 Mandatory Personal Protective Equipment (PPE): Operators must wear certified safety eyewear (ANSI Z87.1), nitrile or heat-resistant protective gloves, cleanroom lab coat, and steel-toe non-slip footwear.

4.2 Safety Interlocks: Verify all physical lid latches, safety interlocks, and emergency stop mechanisms are 100% functional prior to initiating power.

4.3 Hazard Containment: In the event of abnormal vibration, acoustic resonance, or fluid leakage, press the Emergency Stop button immediately and notify the Shift Supervisor.

5.0 Materials, Reagents, and Equipment Specifications

5.1 Calibrated Production Machinery with digital controller interface.

5.2 Standard Reagents & Cleaning Solutions: 70% v/v Isopropyl Alcohol (IPA) USP grade, sterile deionized water (DIW).

5.3 Calibrated Instrumentation: Digital tachometer, thermocouple thermometer, and certified pressure gauge with valid calibration stickers.

6.0 Operating Procedure

6.1 Line Clearance & Inspection: Verify equipment is clean, dry, free of residual matter, and valid calibration tags are affixed.

6.2 Pre-Operation Verification: Energize main power switch. Confirm digital display illuminates without fault codes.

6.3 ****Parameter Configuration****: Program the primary operating cycle:

- Operating Speed: Set to 3000 RPM $\pm$ 50 RPM.
- Operating Temperature: Maintain at 20.0 °C $\pm$ 2.0 °C.
- Operating Chamber Pressure: Regulate at 15 PSI $\pm$ 1 PSI.

6.4 ****Cycle Initiation****: Close and lock the chamber safety lid. Verify magnetic interlock engages. Press 'START' on the controller panel.

6.5 In-Cycle Monitoring: Continuously monitor operational readouts for duration of batch run. Do NOT attempt to open or bypass interlocks while moving parts are active.

6.6 ****Cycle Completion****: Allow system to come to a complete zero-energy standstill. Verify digital readout displays 0 before disengaging lid latch.

7.0 In-Process Quality Checks and Acceptance Criteria

7.1 Speed Verification: Operating speed must remain within $\pm$ 1.5% of setpoint throughout duration.

7.2 Thermal Stability: Chamber temperature must stay within $\pm$ 0.5 °C of target specification.

7.3 Visual Inspection: Finished product must meet clear physical appearance criteria with zero foreign particulate matter.

7.4 Supervisory Verification: The Shift Supervisor must sign off on the batch execution log prior to line handover.

8.0 Troubleshooting and Corrective Actions

8.1 Fault Code E-01 (Speed Deviation): Halt cycle, check drive belt tension and rotor balance. If fault persists, tag out machine and contact Maintenance.

8.2 Fault Code E-04 (Thermal Excursion): Verify coolant lines and refrigerant pressure. Ensure ambient room temperature is within 18°C - 24°C.

8.3 Unplanned Stoppage: Document stoppage timestamp, total elapsed cycle time, and initiate an unplanned event record in accordance with QA Deviation SOP.

9.0 Documentation, Record Retention, and Data Integrity

9.1 Real-Time Documentation: All operating parameters, timestamps, lot numbers, and technician initials must be recorded contemporaneously in the batch production record.

9.2 Record Archival: Completed batch logs and calibration records must be archived in the site document repository for a minimum retention period of seven (7) years.

9.3 Data Integrity: No blank fields, correction fluid, or unverified alterations are permitted per ALCOA+ standards.

10.0 Revision History and Document Approval

10.1 Version 1.0 (Harmonized Release): Initial consolidation of legacy procedures (SOP-Centrifuge-MethodA.pdf, SOP-Centrifuge-MethodB.pdf) into standard Kenvue 10-tier procedure format.

10.2 Author: Operations Harmonization Working Group | Date: October 2026

10.3 Technical Reviewer: Quality Operations Lead | Status: Approved

10.4 Quality Approver: Head of Site Quality & Compliance | Status: Effective